The most optimistic of string theorists envisioned that these differences would serve to eliminate four of the five versions when detailed comparisons to experimental data could one day be carried out. But, frankly, the mere existence of five different formulations of string theory was a source of quiet discomfort. The dream of unification is one in which scientists are led to a unique theory of the universe. If research established that only one theoretical framework could embrace both quantum mechanics and general relativity, theorists would reach unification nirvana. They would have a strong case for the framework's validity even in the absence of direct experimental verification. After all, a wealth of experimental support for both quantum mechanics and general relativity already exists, and it seems plain as day that the laws governing the universe should be mutually compatible. If a particular theory were the unique, mathematically consistent arch spanning the two experimentally confirmed pillars of twentieth-century physics, that would provide powerful, albeit indirect, evidence for the theory's inevitability.

But the fact that there are five versions of string theory, superficially similar yet distinct in detail, would seem to mean that string theory fails the uniqueness test. Even if the optimists are some day vindicated and only one of the five string theories is confirmed experimentally, we would still be vexed by the nagging question of why there are four other consistent formulations. Would the other four simply be mathematical curiosities? Would they have any significance for the physical world? Might their existence be the tip of a theoretical iceberg in which clever scientists would subsequently show that there are actually five other versions, or six, or seven, or perhaps even an endless number of distinct mathematical variations on a theme of strings?

During the late 1980s and early 1990s, with many physicists hotly pursuing an understanding of one or another of the string theories, the enigma of the five versions was not a problem researchers typically dealt with on a day-to-day basis. Instead, it was one of those quiet questions that everyone assumed would be addressed in the distant future, when the understanding of each individual string theory had become significantly more refined.